

1-D:

Net social benefits are calculated by subtracting social costs from social benefits, or in other words, $SB - SC$.

2-D:

D) In a free market, private builders are motivated by profit and tend to ignore external costs, such as noise and disturbance that are not directly borne by them. This is known as a negative externality. **Since the builders don't have to pay for these external costs, they don't take them into account when making decisions about the building project.**

A) Education is not a public good since it exhibits the characteristics of excludability and rivalry.

B) The number of people objecting to the building is not the key issue; it's the fact that the builders don't bear the costs of the noise and disturbance.

C) The government's belief about private school construction is not directly relevant to the market failure to account for external costs.

3-C:

A - When average cost is equal to average revenue, this represents normal profits.

B – When Average cost is equal to marginal revenue, this will not ensure allocative efficiency.

C – For allocative efficiency to be achieved, marginal cost must be equal to average revenue (price), which is why this becomes correct.

D – When $MC = MR$, this represents profit maximizing level of output.

4-C:

A – If there was an increase in consumer's money income, there must have been an increase in the quantities consumed of both goods.

B – If there was a decrease in consumer's real income, there must have been a decrease in the quantities consumed of both goods.

C – If good Y became relatively expensive, there will be a contraction in quantity demanded of good Y. And since good X now becomes relatively cheaper, there will be an extension in qty demanded of good X. This is why, this becomes correct

D – If the price of good X increased, then in that case, the quantity of good X may have decreased.

5-C:

① Normal goods - positive

price falls, qty extends

substitution effect = demand change

price falls, real income rises and qty demanded extends

$P \downarrow, Y \uparrow, Q \uparrow \Rightarrow$ income effect $\Rightarrow$ demand changes $\Rightarrow$ positive

② Normal good

price rises, qty contracts

$P \uparrow, Q \downarrow$

substitution effect = negative

price rises, real income falls, qty demanded contracts

$P \uparrow, Y \downarrow, Q \downarrow \Rightarrow$ income effect = negative

Interior goods:

A) price falls, qty extends

substitution effect: change in demand = positive

Income effect: price falls, real income rises, qty contracts

$P \downarrow, Y \uparrow, Q \downarrow$

change in demand = negative

Interior goods: $P \uparrow, Q \downarrow$

B) price rises, qty contracts

substitution effect: change in demand = negative

Income effect: price rises, real income falls, quantity extends $P \uparrow, Y \downarrow, Q \uparrow$

change in demand: positive

This explains why key C will be correct.

6-D: Since, the government has imposed a **per-unit tax of \$10** on each product, the average cost will increase such that the curve will shift upwards "parallelly".

- The reason behind a parallel shift can be seen in detail below.

A - It can be seen below that the average cost rises by \$10.

B -

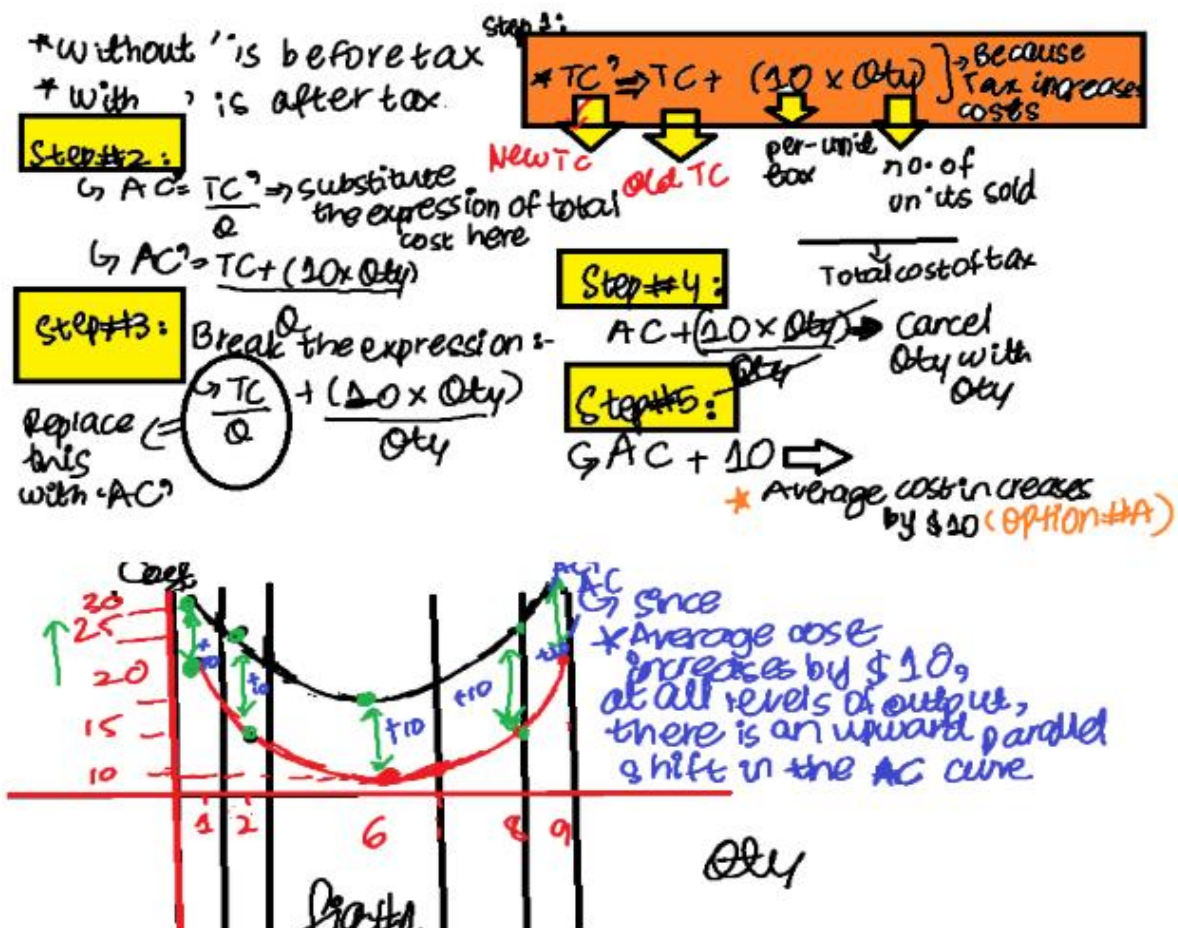
B - The marginal cost will change too, this can be seen in the table below.

	AC				AC'		
AC	20	15	20		30	25	20
Qty	1	2	6		1	2	6
TC	20	30	60		30	50	120
MC	20	10	30		30	20	70

C : Since the average cost curve shifts upwards parallelly, the minimum average cost output will be unchanged at the quantity of "6 units"

D : Since the average cost curve shifts upwards parallelly, the minimum average cost output will be unchanged at the quantity of "6 units". This is why, key D, is correct.

The numerical example explains this in detail



7-B:

A – In a non-collusive oligopoly market structure, due to a high degree of interdependence, firms will match the price cuts of rivals, to prevent the firm who decreased the price, from capturing their market shares.

B – In a non-collusive oligopoly, if one firm increases the price in a desire to earn higher revenues, other rivals will not respond to it, and so, the firm who raised the price will face a fall in the quantity demanded by a relatively higher percentage and subsequently Total revenues will decrease too. This explains why B is correct.

C – In a non-collusive oligopoly, product differentiation occurs as these firms compete with each other by means of non-price competition.

D – Large firms for example firms in an oligopoly do earn supernormal profits in both , the short run and the long-run.

8-D:

→ Since, total cost is only **based on “fixed cost” here**, which does not change despite changes in the level of production, marginal cost will be “zero”
When Marginal cost (which is zero) is equal to Marginal revenue(which is zero), since the condition of $MC=MR$ is fulfilled , this means that the monopolist’s **profit-maximizing price will be where Marginal Revenue is equal to zero, and when $MR=0$, the equilibrium price will be “D”.**

This is why, key D, will be correct.

A – Here, $ATC > AR$, which represents the fact that “losses” are being made.

B – Here, $AR=ATC$, which means that “normal profits” will be made.

C – Here, $MR=ATC$, which would not maximize profits.

9-A:

A - If R (steel) takes over S(ships) industry, this an example of forward vertical integration because **steel is a raw material of the ships industry**. This is why, key A, becomes correct.

B- If S(ships) takes over V(cruise liners), this is not an example of backward-vertical integration. This is because backward vertical integration involves a company merging with or acquiring a supplier.

C- If T(oil tanker) takes over S(ships), this is an example of “forward-vertical integration” .If Company T, which operates oil tankers, were to take over Company S, which produces ships, it would indeed be an example of **forward vertical integration**. In this scenario, Company T acquiring Company S represents **forward vertical integration** because Company T is acquiring a company involved in the production of ships, which will enable oil tankers to be sold abroad through ships.

D – If cruise liners company(company V) takes over ships manufacturing company(company S), this is not an example of horizontal integration as firstly, they are not operating in the same industry, and secondly, they are also not operating at the same stage of production.

10-A:

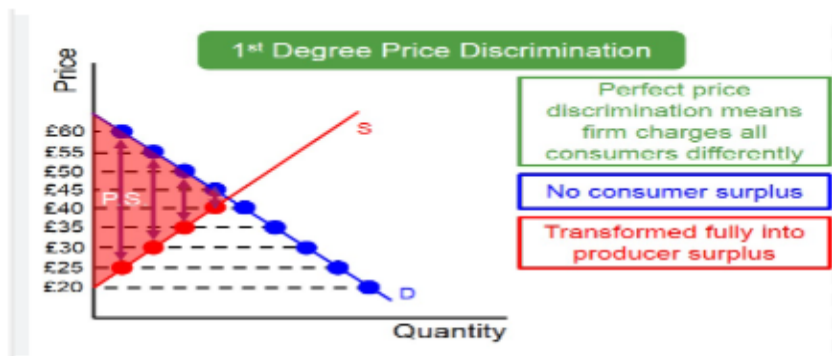
A – Normal profits will be made when Average revenue = Average total cost. This is why, key A, becomes correct.

B – Average revenue is only equal to marginal revenue in case of perfect competition.

C – Where marginal cost is at its minimum, normal profits would not occur, because normal profits occur when $AR=AC$. **Importantly, there is no such equilibrium where the firm operates at the minimum of Marginal cost.**

D – The monopolist will be allocatively efficient when $MC=AR(\text{price})$, and here, normal profits would not be made.

11-C:



As can be seen above, option C, is correct because “**first degree price discrimination**” occurs when prices are determined by the maximum price each consumer is willing to pay.

A - A kinked demand curve is a concept used in oligopoly theory to explain **price rigidity** and stability in markets with a small number of **interdependent firms**. It suggests that firms in an oligopolistic market may face a discontinuity or kink in their perceived demand curve at the current market price.

B- Limit pricing refers to the practice of setting prices at a level that is low enough to discourage potential entrants from entering the market.

D - Price leadership occurs when one dominant firm in an industry sets the price, and other firms follow suit. The scenario does not indicate any firm setting the price and others adjusting accordingly. This occurs mainly in case of a **dominant oligopoly**.

12-B:

A) This option is incorrect. Allowing companies to take over other companies does not directly affect their ability to borrow from the banking system. Companies can already borrow from banks and other financial institutions regardless of merging.

B) This is the correct answer. Allowing companies to take over other companies can lead to productive efficiency gains through economies of scale, elimination of redundant resources, and sharing of best practices which will bring about a reduction in costs and increased competitiveness.

C) This option is incorrect. Mergers and acquisitions do not necessarily lead to rising wage rates. In fact, they may sometimes result in job losses, especially if the acquiring company seeks to eliminate redundant positions or reduce costs.

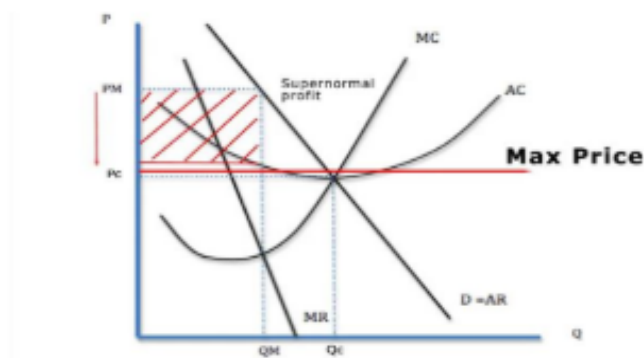
D) This option is incorrect. While investment bankers may benefit from mergers and acquisitions deals, this is not a primary argument for allowing companies to take over others.

13-C:

A – It will not encourage electricity suppliers to invest in additional capacity because , as price falls due to the maximum price legislation, quantity supplied contracts.

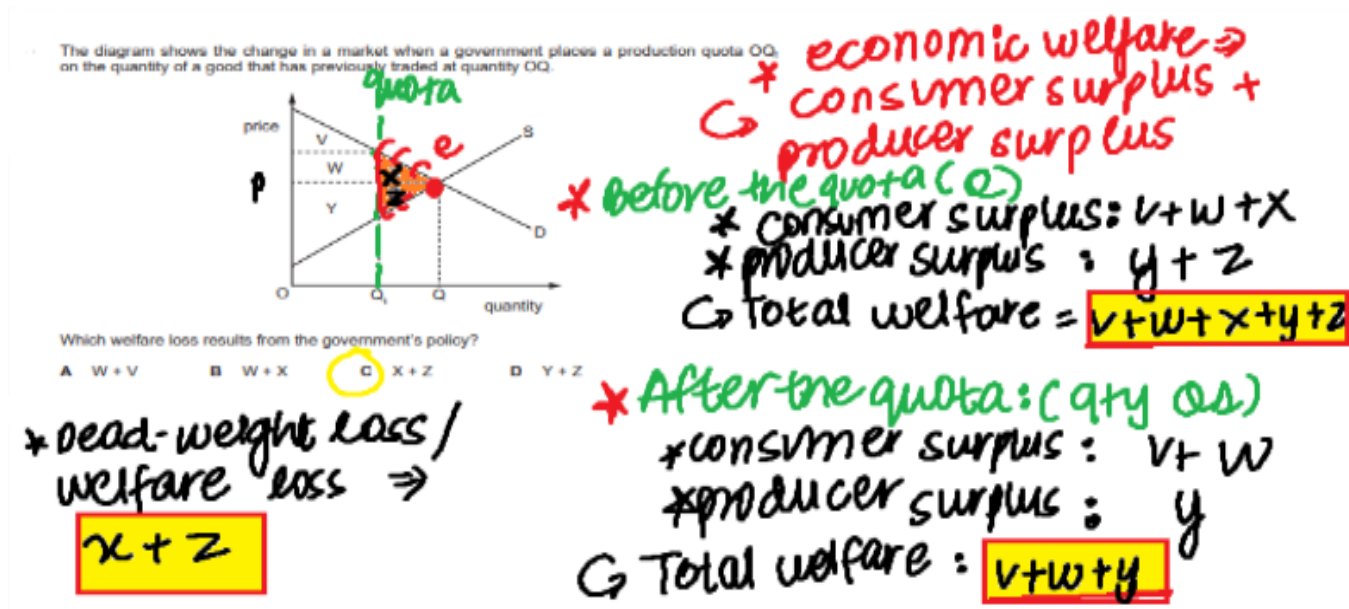
B – As price falls due to the maximum price legislation, the quantity demanded extends, which means that, energy is not being conserved.

C – Since, it is a regulation through which prices are fixed and lowered to below the equilibrium, it will prevent consumers from being exploited by a “monopoly”. This is why, this becomes correct. This can be seen in the diagram below:



D – Because of a maximum price legislation, when prices are lowered, this results in a contraction in qty supplied and an extension in qty demanded, which leads to a “shortage”, and to overcome this, **rationing** will have to be done by the government.

14-C:



15-C:

- A- A ban on cigarette advertising will lead to a fall in the demand for cigarettes, as a result of which, the external costs incurred would reduce.
- B- A ban on cigarette smoking in public areas will again cause the demand for cigarettes to fall, as a result of which, the external costs incurred would reduce.
- C- A ban on substitutes for cigarettes will lead to consumers switching to cigarettes, as a result of which, the demand for cigarettes will rise, and so, the negative externalities will increase. This is why, this becomes correct.
- D- A ban on sale of cigarettes will lead to a fall in the supply of cigarettes, as a result of which, the external costs incurred would reduce.

16-A:

A - Since, a teacher would be willing to continue to do the job if she earned at-least \$25000, this represents that this amount represents the "transfer earnings"

And since, she is currently earning \$30,000, this represents that \$5000 is the "economic rent" as a teacher. This is why, key A, becomes correct.

B – Her economic rent as an accountant is \$15000 (\$40,000-\$25000)

C and D– Her transfer earnings as both: a teacher and an accountant will be \$25000

17-B:

A – If the economy is experiencing a fall in the price of capital, in that case, the demand for labor will fall, which means that employer will substitute labor with capital, therefore resulting in a fall in the wage rate, accompanied with a rise in unemployment.

B – If the economy is experiencing rising employment, this means that the purchasing powers of individuals will rise, as a result of which, there will be a rise in the demand of products produced by the industry, and so, there will be a rise in the demand for labor, which is a derived demand. The effect of this will be to raise wages and employment simultaneously. This is why, key B, becomes correct.

C – If the price elasticity of the industry's goods is =1, or in other words, unitary, this will not lead to a rise in wage rate alongside a rise in employment.

D - If the demand for the product is elastic, then a firm cannot pass a greater burden of a rise in costs as a result of: for example, a rise in wages to consumers, as a result of which, the demand for labor, which is the "derived demand" will be elastic too. Now, upon a wage rise, when costs rise, and even when the firm raises prices by a relatively smaller percentage, keeping a higher percentage of the cost burden to itself, quantity demanded will contract by a relatively higher percentage, resulting in a fall in total revenue. Now, the firm will lay back workers to maintain profitability, thus increasing the level of "unemployment".

18-B:

The table shows the number of workers employed by four firms in the same industry, the average number of hours worked each day and the total output each firm produces each day.

Which firm's workers have the highest productivity?

Total no of hours worked	number of workers	average number of hours worked per day	total output per day (units)	Total output / no of hours	Ans
32	4	X	320	320/32 → 10	
35	5	X	420	420/35 → 12 *	
42	6	X	462	462/42 → 11	
63	7	X	630	630/63 → 10	

Total no of hours worked = no of workers × average no of hours worked.

Highest productivity
→ option B

This is why, key B, will be correct.

19-B:

The correct answer is B) recession.

The business cycle, also known as the trade cycle, refers to the fluctuations in economic activity, typically involving periods of expansion and contraction. The output gap is the difference between an economy's actual output and its potential output.

During a recession (B), the economy is in a contraction phase, and the output gap is likely to be negative, indicating that the economy is producing less than its potential output. **As the recession deepens, the negative output gap will increase.**

A) Boom: During a boom, the economy has expanded significantly and the output gap is likely to be positive.

C) Recovery: In the recovery phase, the economy is transitioning from a recession to expansion, and the negative output gap is likely to be decreasing

D) Trough: The trough is the lowest point of a recession, and while the output gap is likely to be negative, it is not increasing (as the economy has already contracted and is preparing for recovery).

20-B:

Dominant sector of industry: Primary sector

In the least developed countries (LDCs), the primary sector is the **dominant sector of industry**. This includes agriculture, mining, and extractive industries, which are the main sources of employment, income, and exports. This means that LDCs rely heavily on basic industries and have limited industrial development, making them vulnerable to external shocks and fluctuations in global commodity prices.

Export earnings: Unstable

The export earnings of LDCs are often unstable due to various factors. These countries rely heavily on exports of primary commodities, such as agricultural products, minerals, and raw materials. However, the prices of these commodities can fluctuate significantly due to changes in global demand and supply. Additionally, LDCs have limited market access and often face trade barriers, making it difficult for them to maintain consistent export earnings.

Terms of trade: Unstable over time

The terms of trade for LDCs are often unstable over time, meaning that the ratio of export prices to import prices can fluctuate significantly. This is due to various factors, including changes in global commodity prices, fluctuations in exchange rates, and differences in inflation rates between LDCs and their trading partners. **As a result, LDCs often face unfavorable terms of trade, meaning they receive low prices for their exports and pay high prices for imports. This can make it difficult for LDCs to maintain a stable trade balance, invest in development, and reduce poverty.**

21-B:

Economic growth refers to **an increase in the production of goods and services in an economy, often measured by an increase in GDP**. While economic growth is central to economic development, the two concepts are not identical. **Economic development is a broader concept that encompasses not only economic growth but also improvements in living standards, health, education, and overall well-being.**

A) An expansion of the range of economic choices available to individuals will increase standard of living of individuals and is central to economic development,

C) The eradication of absolute poverty is a key aspect of economic development, as it ensures that basic needs are met and individuals have access to essential goods and services.

D) The satisfaction of basic needs amongst the whole population is also central to economic development, as it ensures that everyone has access to necessities like food, water and shelter.

In contrast, an increase in GDP (option B) is central to economic growth but not necessarily to economic development, as it only measures the total value of goods and services produced, without considering important aspects like income distribution, poverty reduction, and human development.

22-D:

Let's say:

Pakistan's GNP per head is PKR 500,000

The example is based on the assumption that the actual market ER is USD 1 = PKR 200.

Using market exchange rates, Pakistan's GNP per head in USD would be:

$$\text{PKR } 500,000 \div 200 = \text{USD } 2,500$$

Now, let's assume the same GNP per head in the US is USD 50,000.

Based on this comparison, it seems like the US has a much higher living standard (20 times higher).

But, what if the same PKR 500,000 in Pakistan can buy more goods and services than USD 2,500 can in the US? This is where Purchasing Power Parity (PPP) comes in.

Let's say the PPP rate or the real er is PKR 100 = USD 1 (meaning PKR 100 in Pakistan has the same purchasing power as USD 1 in the US).

Using PPP, Pakistan's GNP per head in USD would be:

$$\text{PKR } 500,000 \div 100 = \text{USD } 5,000$$

Now, the comparison with the US GNP per head (USD 50,000) is more realistic, and the difference in living standards is not as drastic (10 times higher instead of 20).

This example illustrates how using market exchange rates can lead to an incorrect ranking, and why PPP is a more appropriate method for comparing living standards between countries. The difference between the market exchange rate and the PPP rate comes from the variations in the cost of living between countries.

In our example, the PKR 500,000 can buy more goods and services in Pakistan than the equivalent USD 2,500 can in the US, because the cost of living in Pakistan is lower than in the US. **This means that it would be the most essential element in listing countries in order of their standards of living during a particular year.**

23-D:

A - If greater health and safety rules are established for manufacturers, this will be most likely to increase the costs of employment as a result of which, employers are most likely to reduce the demand for labor, which will increase unemployment.

B - Increasing income tax would reduce disposable income, leading to lower consumption and aggregate demand, and when production and demand for labor falls, this will exacerbate unemployment.

C) Increasing the rate of interest would make borrowing more expensive, reduce investment, and decrease aggregate demand, and when because of this production and demand for labor falls, unemployment rises.

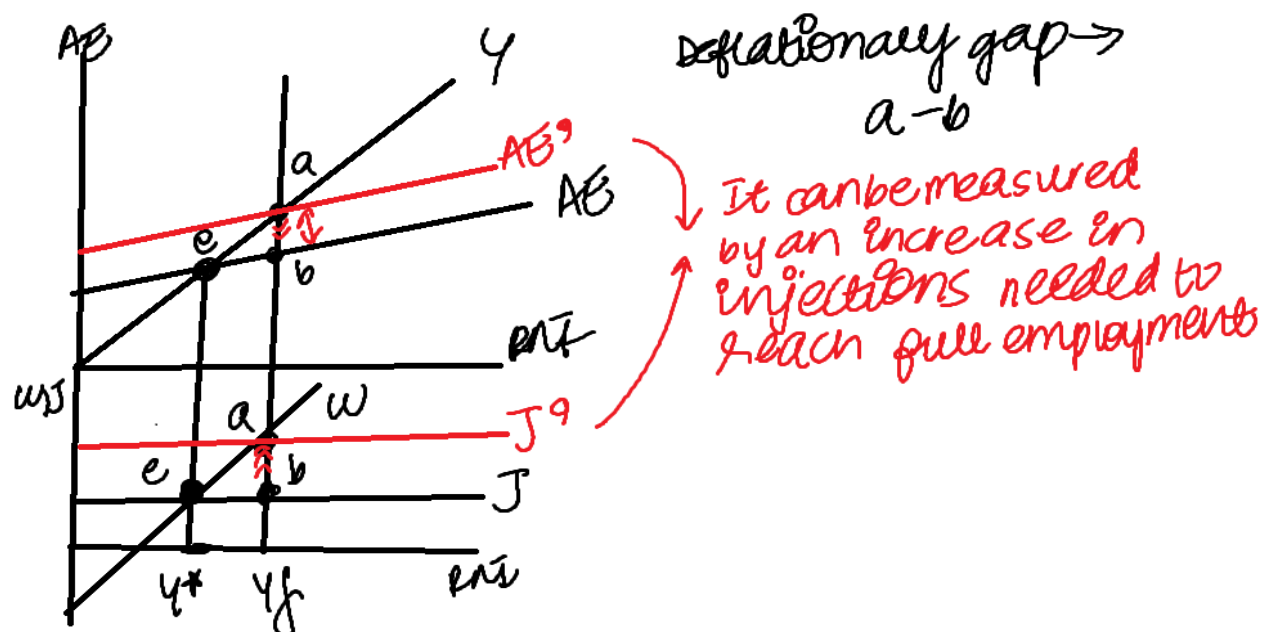
D – Subsidizing the building of houses represents an increase in government spending which will increase Aggregate demand. This will increase production and demand for labor which will reduce unemployment. This is why, key D, becomes correct.

24-B:

A,C and D: Increases in government spending on anti-smoking campaigns, government spending in form of wages of government employees represent injections into the circular flow since they will directly increase Aggregate demand. However, transfer payments may not increase injections because they don't contribute to GDP.

B : Similarly, an increase in imports of luxury cars represent leakages from the circular flow.

25-A:



This is why, only option A can be correct. Public sector borrowing requirement, the rate of deflation and the rate of unemployment cannot be used to measure deflationary gap.

26-D:

Marginal propensity to consume represents the proportion of additional income that is consumed. The formula is: Change in consumption/Change in income.

26 The following data describes an economy's consumption function.

	income (\$m)	consumption (\$m)
a	100	90
	140	122
	180	154
	220	186

Handwritten calculations: $\Delta C = 32$ (between 100 and 140 income), $\Delta Y = 40$ (between 100 and 140 income). $\frac{\Delta C}{\Delta Y} = \frac{32}{40} \Rightarrow 0.8$ (circled in red). An arrow points from the calculation to the table.

What can be deduced about the value of the marginal propensity to consume in this economy?

- A It increases as income increases.
- B It decreases as income increases.
- C It is 0.90.
- ☒ D It is 0.80.

27-A:

A - When a consumer uses an overdraft facility to buy a car, they are essentially borrowing money from the bank to make the purchase. This increases the money supply immediately, as the bank creates new money by crediting the consumer's account. **Only option A involves the creation of new money, leading to an immediate increase in the domestic money supply**

B - Using a reserve of cash to buy a car doesn't increase the money supply, as the cash was already in circulation.

C - When a consumer uses reserves of Euro to buy a car from abroad, this essentially reduces the domestic money supply as money leaks from the economy.

D - Using a current bank account balance to buy a car simply transfers money from one account to another, without increasing the overall money supply.

28-D:

Kuznets curve suggests that as an economy grows, income inequality initially increases, but eventually decreases. The curve has two phases:

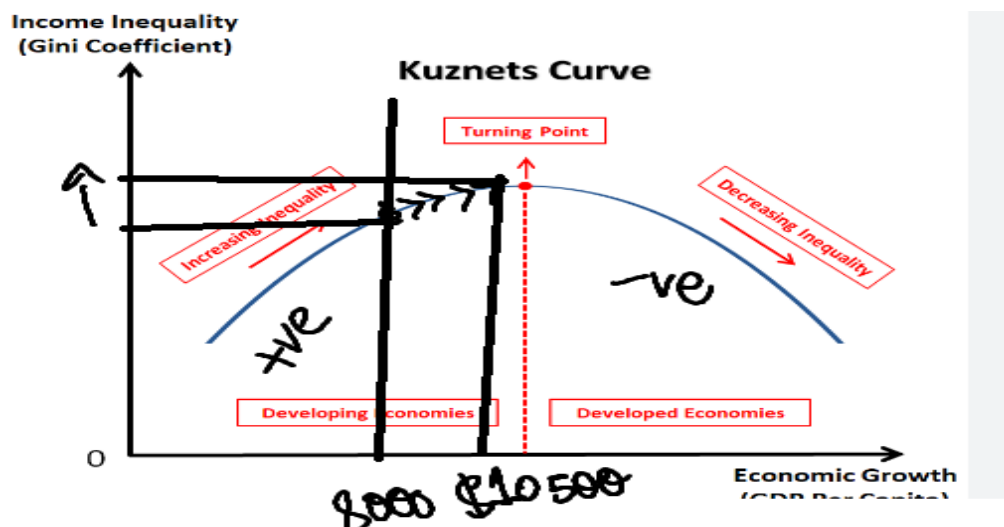
Phase 1: Increasing Inequality (Initial Stage of Economic Growth)

In this phase, economic growth is driven by industrialization and urbanization. As low-income people move from rural areas to cities, their wages are further suppressed, which increases rich poor disparity. The owners of capital and natural resources accumulate wealth, widening the income gap. Skilled workers and

entrepreneurs tend to benefit more from economic growth, while unskilled workers are left behind. Urbanization leads to a concentration of wealth in cities, leaving rural areas behind.

Phase 2: Decreasing Inequality (Later Stage of Economic Growth)

In this phase, economic growth becomes more inclusive, and income inequality starts to decline. Governments implement policies to reduce inequality, such as progressive taxation, social security, and education. The middle class expands, and a larger share of the population benefits from economic growth. Technological progress and innovation reduce the concentration of wealth, making it more accessible to a broader population. Overall, the Kuznets Curve suggests that economic growth initially widens income inequality but eventually leads to a more equitable distribution of wealth..



The graph above explains why only option D will be correct.

29-B:

The correct answer is B) high growth and low unemployment.

High economic growth and low unemployment are complementary macroeconomic policy aims. Economic growth arising from an increase in Aggregate demand brings about an increase in production, demand for labor and employment, reducing unemployment.

A) High growth and low inflation may not always go together, as rapid growth can lead to inflationary pressures. For example, a rapidly growing economy may

experience increased demand for goods and services, which means that because of this Aggregate demand may outpace Aggregate supply, and the resultant shortages can translate into higher prices and subsequently demand pull inflation.

C) Low inflation and low unemployment are not directly related, as usually there is a negative relationship between inflation and unemployment as outlined through the Philips curve. This is because a lower degree of unemployment means that people have high incomes, and when because of this consumption and Aggregate demand increases, there may come a point where Aggregate demand may outpace Aggregate supply, and the resultant shortages can translate into higher prices and subsequently demand pull inflation.

D) Low unemployment and a balance of trade surplus are not directly complementary. This is because, when levels of unemployment decrease, this represents that people have high incomes as a result of which, demand for imported goods may increase, and a rise in import expenditures will result in a move towards a Balance of trade deficit.

30-C:

A - A fall in the country's rate of inflation would make domestic investments more attractive due to the possibility of anticipated returns , leading to an inflow of capital funds.

B – An expected appreciation of the country's exchange rate means that we can expect cheaper imports and an increase in import expenditures. But since in this question we are interested about an outflow of capital funds, this becomes incorrect.

C - An increase in foreign interest rates would make foreign investments more attractive, leading to an outflow of capital funds from the country as investors seek higher returns abroad. This is why, option C is correct.

D – An increase in foreign prices represent that foreign countries are facing a higher rate of inflation. This would make domestic investments more attractive due to the possibility of anticipated returns , leading to an inflow of capital funds.

